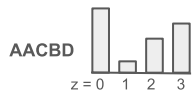
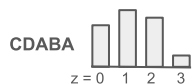
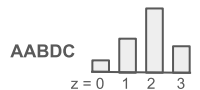


Counterfactual grouping intervention

Original motif identities
and sequence-dependent
affinity profiles a_{mz}



Local-state class z
 $z \in \{0, 1, 2, 3\}$

Symbolic affinity profiles;
(not quantitative)

Grouping map

$g: \mathcal{M} \rightarrow \{1, \dots, K_g\}$
Grouping map assigns
motif identities to groups

group $g(m) = 1$

AABDC
CDABA

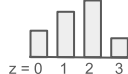
group $g(m) = 2$

AACBD
DABCD

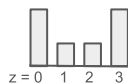
Group-average replacement

Group-average affinity profile

group $g(m) = 1$



group $g(m) = 2$



Within-group affinity-profile
distinctions removed

Between group affinity-profile
distinctions remain

Intervened motif identities
and assigned profiles

AABDC $\rightarrow$ group 1 average profile
CDABA $\rightarrow$ group 1 average profile
AACBD $\rightarrow$ group 2 average profile
DABCD $\rightarrow$ group 2 average profile

Changed: sequence-dependent
affinity profiles used for local-state
assignment

Unchanged:
starting population
motif identities
positional segments
adjacency categories
population-update rule

Intervention families:
balanced-random grouping
affinity-rank grouping
contiguous-substring grouping

Constant endpoint

All motif identities in one group ($K_g = 1$)

One global average affinity profile
assigned to every motif identity

$$I(g(M); Z | S) = 0$$

Identity endpoint

Every motif identity remains ($K_g = |\mathcal{M}|$)

Each singleton group retains
its original affinity profile

$$I(g(M); Z | S) = I(M; Z | S)$$

Actual retained information

$$I_{ret,r}(g) = I(g(M); Z | S)$$

Measured for the specific
grouping map g

Not inferred from K_g , nominal group
count, or a logarithmic resolution heuristic